Maximum Likelihood for a Bernoulli Model

Last modified: 2026-10-21 14:06:33 (PDT)

Maximum likelihood for a Bernoulli model without covariates

The likelihood

Exercise 0.1 (Likelihood of binary outcomes with one event probability)**.** A binary outcome Y with event probability π has

$$\begin{aligned}
&\Pr(Y = 1) = \pi && \text{(definition of the event probability)}\\
&\Pr(Y = 0) = 1 - \pi && \text{(complement rule)}\\
&\Pr(Y = y) = \pi^y (1-\pi)^{1-y}, \quad y \in \{0, 1\} && \text{(combine the two cases)}
\end{aligned}$$

Let $\underset{\sim}{y} \stackrel{\text{def}}{=} (y_1, \ldots, y_n)$ be a dataset of mutually independent binary outcomes, all with the same event probability π: $Y_i \sim_{\perp\!\!\!\perp} \text{Ber}(\pi)$. Derive the likelihood of $\underset{\sim}{y}$.

Solution

Solution 0.1.

$$
\begin{aligned}
\mathcal{L}(\pi; \underset{\sim}{y}) &= \prod_{i=1}^{n} \mathcal{L}_i(\pi) && \text{(product of likelihood components)} \\
&= \prod_{i=1}^{n} \Pr(Y_i = y_i) && \text{(definition of likelihood components)} \\
&= \prod_{i=1}^{n} \pi^{y_i}(1-\pi)^{1-y_i} && \text{(Bernoulli PMF)} \\
&= \left(\prod_{i=1}^{n} \pi^{y_i}\right)\left(\prod_{i=1}^{n}(1-\pi)^{1-y_i}\right) && \text{(regroup the factors of the product)} \\
&= \pi^{\sum_{i=1}^{n} y_i}(1-\pi)^{\sum_{i=1}^{n}(1-y_i)} && \text{(product of powers of a common base)} \\
&= \pi^{\sum_{i=1}^{n} y_i}(1-\pi)^{\sum_{i=1}^{n} 1 - \sum_{i=1}^{n} y_i} && \text{(split the sum in the exponent)} \\
&= \pi^{\sum_{i=1}^{n} y_i}(1-\pi)^{n - \sum_{i=1}^{n} y_i} && \left(\textstyle\sum_{i=1}^{n} 1 = n\right)
\end{aligned}
$$

Theorem 0.1 (Likelihood of binary outcomes with one event probability)**.** *Let $Y_1, \ldots, Y_n$ be mutually independent binary random variables, each with the same event probability π: $Y_i \sim_{\perp\!\!\!\perp} \text{Ber}(\pi)$. For observed outcomes $\underset{\sim}{y} \overset{def}{=} (y_1, \ldots, y_n)$, with each $y_i \in \{0, 1\}$, the likelihood $\mathcal{L}(\pi; \underset{\sim}{y})$ of π depends on the data only through the number of events $\sum_{i=1}^{n} y_i$:*

$$\mathcal{L}(\pi; \underset{\sim}{y}) = \pi^{\sum_{i=1}^{n} y_i}(1-\pi)^{n - \sum_{i=1}^{n} y_i}$$

Proof

Proof. This is the solution to Exercise 0.1, worked in Solution 0.1. □

Log-likelihood contributions

Exercise 0.2 (Log-likelihood contribution of one binary outcome)**.** Let:

- $Y_i \sim_{\perp\!\!\!\perp} \text{Ber}(\pi)$ for $i = 1, \ldots, n$, with no covariates;
- $y_i \in \{0, 1\}$ be the observed value of Y_i.

Derive the log-likelihood contribution $\ell_i(\pi)$ of observation i, the log of its likelihood component.

Solution

Solution 0.2.

$$\begin{aligned}
\ell_i(\pi) &= \log\{\Pr(Y_i = y_i)\} && \text{(log of the likelihood component)} \\
&= \log\{\pi^{y_i}(1-\pi)^{1-y_i}\} && \text{(Bernoulli PMF)} \\
&= \log\{\pi^{y_i}\} + \log\{(1-\pi)^{1-y_i}\} && \text{(log of a product)} \\
&= y_i \log\{\pi\} + (1-y_i)\log\{1-\pi\} && \text{(log of a power)}
\end{aligned}$$

Theorem 0.2 (Log-likelihood contribution of one binary outcome)**.** *If* $Y_i \sim \text{Ber}(\pi)$ *has observed value* $y_i \in \{0, 1\}$*, its log-likelihood contribution is:*

$$\ell_i(\pi) = y_i \log\{\pi\} + (1-y_i)\log\{1-\pi\}$$

Proof

Proof. This is the solution to Exercise 0.2, worked in Solution 0.2. Rewriting it with the log-odds $\text{logit}(\pi)$ and summing over i gives Theorem 0.3. □

The log-likelihood of the sample

Exercise 0.3 (Log-likelihood of binary outcomes with one event probability)**.** Derive the log-likelihood of $\underset{\sim}{y}$ from Exercise 0.1.

Solution

Solution 0.3. Starting from the likelihood in Theorem 0.1:

$$
\begin{aligned}
\ell(\pi; \underset{\sim}{y}) &= \log\Big\{\pi^{\sum_{i=1}^n y_i}(1-\pi)^{n-\sum_{i=1}^n y_i}\Big\} && \text{(log of the likelihood)} \\
&= \log\Big\{\pi^{\sum_{i=1}^n y_i}\Big\} + \log\Big\{(1-\pi)^{n-\sum_{i=1}^n y_i}\Big\} && \text{(log of a product)} \\
&= \left(\sum_{i=1}^n y_i\right)\log\{\pi\} + \left(n - \sum_{i=1}^n y_i\right)\log\{1-\pi\} && \text{(log of a power)} \\
&= \left(\sum_{i=1}^n y_i\right)\log\{\pi\} + n\log\{1-\pi\} - \left(\sum_{i=1}^n y_i\right)\log\{1-\pi\} && \text{(distribute } \log\{1-\pi\}\text{)} \\
&= \left(\sum_{i=1}^n y_i\right)\log\{\pi\} - \left(\sum_{i=1}^n y_i\right)\log\{1-\pi\} + n\log\{1-\pi\} && \text{(reorder the terms)} \\
&= \left(\sum_{i=1}^n y_i\right)(\log\{\pi\} - \log\{1-\pi\}) + n\log\{1-\pi\} && \text{(collect the terms in } \textstyle\sum_{i=1}^n y_i\text{)} \\
&= \left(\sum_{i=1}^n y_i\right)\log\left\{\frac{\pi}{1-\pi}\right\} + n\log\{1-\pi\} && \text{(log of a quotient)} \\
&= \left(\sum_{i=1}^n y_i\right)\text{logit}(\pi) + n\log\{1-\pi\} && \text{(definition of logit)}
\end{aligned}
$$

Theorem 0.3 (Log-likelihood of binary outcomes with one event probability)**.** *Let* $Y_i \sim_{\perp\!\!\!\perp} \text{Ber}(\pi)$ *for* $i = 1, \ldots, n$*, with observed outcomes* $y \overset{def}{=} (y_1, \ldots, y_n)$*, each* $y_i \in \{0, 1\}$*. The log-likelihood* $\ell(\pi; \underset{\sim}{y})$ *of the event probability* π*, written with the log-odds* $\text{logit}(\pi)$*, is:*

$$
\ell(\pi; \underset{\sim}{y}) = \left(\sum_{i=1}^n y_i\right)\text{logit}(\pi) + n\log\{1-\pi\}
$$

Proof

Proof. This is the solution to Exercise 0.3, worked in Solution 0.3. It builds on Theorem 0.1. □

The score of one observation

Exercise 0.4 (Score function of a Bernoulli variable)**.** Derive the score function for a single Bernoulli random variable X. In other words, differentiate the log-likelihood of a single Bernoulli random variable X with respect to the event probability parameter π. Simplify as much as possible.

Solution

Solution 0.4. With $n = 1$, Theorem 0.3 gives $\ell = x\log\{\pi\} + (1-x)\log\{1-\pi\}$, so:

$$\begin{aligned}
\ell' &\stackrel{\text{def}}{=} \frac{\partial}{\partial\pi}\ell && \text{(definition of the score)} \\
&= \frac{\partial}{\partial\pi}(x\log\{\pi\} + (1-x)\log\{1-\pi\}) && \text{(Bernoulli log-likelihood)} \\
&= \frac{\partial}{\partial\pi}(x\log\{\pi\}) + \frac{\partial}{\partial\pi}((1-x)\log\{1-\pi\}) && \text{(linearity of differentiation)} \\
&= x\frac{\partial}{\partial\pi}\log\{\pi\} + (1-x)\frac{\partial}{\partial\pi}\log\{1-\pi\} && \text{(constant multiple rule)} \\
&= x\frac{1}{\pi} + (1-x)\frac{\partial}{\partial\pi}\log\{1-\pi\} && \text{(derivative of log)} \\
&= x\frac{1}{\pi} + (1-x)\frac{1}{1-\pi}\frac{\partial}{\partial\pi}(1-\pi) && \text{(chain rule)}
\end{aligned}$$

The inner derivative is:

$$\begin{aligned}
\frac{\partial}{\partial\pi}(1-\pi) &= \frac{\partial}{\partial\pi}1 - \frac{\partial}{\partial\pi}\pi && \text{(linearity of differentiation)} \\
&= 0 - \frac{\partial}{\partial\pi}\pi && \text{(derivative of a constant)} \\
&= 0 - 1 && \text{(derivative of } \pi \text{ with respect to itself)} \\
&= -1 && \text{(subtract)}
\end{aligned}$$

Substituting the inner derivative back in:

$$
\begin{aligned}
\ell' &= x\frac{1}{\pi} + (1-x)\frac{1}{1-\pi}(-1) && \text{(inner derivative is } -1\text{)} \\
&= x\frac{1}{\pi} - (1-x)\frac{1}{1-\pi} && \text{(multiply by } -1\text{)} \\
&= \frac{x}{\pi} - \frac{1-x}{1-\pi} && \text{(multiply)} \\
&= \frac{x(1-\pi)}{\pi(1-\pi)} - \frac{(1-x)\pi}{\pi(1-\pi)} && \text{(write both terms over the common denominator)} \\
&= \frac{x(1-\pi) - (1-x)\pi}{\pi(1-\pi)} && \text{(combine the fractions)} \\
&= \frac{x - x\pi - \pi + x\pi}{\pi(1-\pi)} && \text{(expand the numerator)} \\
&= \frac{x-\pi}{\pi(1-\pi)} && \text{(cancel } x\pi\text{)} \\
&= \frac{x - \mathrm{E}[X]}{\pi(1-\pi)} && (\mathrm{E}[X] = \pi) \\
&= \frac{x - \mathrm{E}[X]}{\mathrm{Var}(X)} && (\mathrm{Var}(X) = \pi(1-\pi))
\end{aligned}
$$

Theorem 0.4 (Score function of a Bernoulli variable)**.** *Let* $X \sim \mathrm{Ber}(\pi)$*, with event probability* π *and observed value* $x \in \{0, 1\}$*. The score* ℓ' *of the single observation* x *is the derivative of its log-likelihood* ℓ *with respect to* π*. It equals the deviation of* x *from the mean* $\mathrm{E}[X]$*, divided by the variance* $\mathrm{Var}(X)$*:*

$$
\ell' = \frac{x - \mathrm{E}[X]}{\mathrm{Var}(X)} = \frac{x-\pi}{\pi(1-\pi)}
$$

Proof

Proof. This is the solution to Exercise 0.4, worked in Solution 0.4. It builds on Theorem 0.3, with $n = 1$. □

Score contributions

Exercise 0.5 (Score contribution of one binary outcome)**.** In the setting of Exercise 0.2, let:

- $\tau \overset{\text{def}}{=} 1/\text{Var}(Y_i)$ be the precision of Y_i.

Derive the score contribution $\ell_i'(\pi) \overset{\text{def}}{=} \frac{\partial}{\partial \pi}\ell_i(\pi)$ of observation i, and write it using τ.

Solution

Solution 0.5.

$$\begin{aligned}
\ell_i'(\pi) &= \frac{\partial}{\partial \pi}\ell_i(\pi) && \text{(definition of the score contribution)} \\
&= \frac{y_i - \pi}{\pi(1-\pi)} && \text{(score of a Bernoulli variable, with } x = y_i\text{)} \\
&= \frac{y_i - \pi}{\text{Var}(Y_i)} && (\text{Var}(Y_i) = \pi(1-\pi)) \\
&= \frac{1}{\text{Var}(Y_i)}(y_i - \pi) && \text{(a fraction is its numerator times the reciprocal of its denominator)} \\
&= \tau(y_i - \pi) && \text{(definition of } \tau\text{)}
\end{aligned}$$

Theorem 0.5 (Score contribution of one binary outcome)**.** *If $Y_i \sim \text{Ber}(\pi)$ has observed value $y_i \in \{0, 1\}$ and precision $\tau = 1/\text{Var}(Y_i) = 1/(\pi(1-\pi))$, its score contribution is the precision times the deviation of y_i from its mean π:*

$$\ell_i'(\pi) = \tau(y_i - \pi)$$

The precision τ is the same for every i, because all Y_i share the event probability π.

Proof

Proof. This is the solution to Exercise 0.5, worked in Solution 0.5. □

The score function of the sample

Exercise 0.6 (Score function of a binary sample)**.** In the setting of Exercise 0.5, let $\bar{y} \overset{\text{def}}{=} \frac{1}{n}\sum_{i=1}^{n} y_i$ be the sample mean of the observed outcomes. Assemble the score function $\ell'(\pi)$ of the whole sample from the score contributions.

Solution

Solution 0.6.

$$
\begin{aligned}
\ell'(\pi) &= \sum_{i=1}^{n} \ell_i'(\pi) && \text{(score of an independent sample is the sum of the score contributions)} \\
&= \sum_{i=1}^{n} \tau(y_i - \pi) && \text{(score contribution of a binary outcome)} \\
&= \tau \sum_{i=1}^{n} (y_i - \pi) && (\tau \text{ is the same for every } i) \\
&= \tau \left(\sum_{i=1}^{n} y_i - \sum_{i=1}^{n} \pi \right) && \text{(split the sum)} \\
&= \tau \left(\sum_{i=1}^{n} y_i - n\pi \right) && (\textstyle\sum_{i=1}^{n} \pi = n\pi) \\
&= \tau(n\bar{y} - n\pi) && (\textstyle\sum_{i=1}^{n} y_i = n\bar{y}) \\
&= n\tau(\bar{y} - \pi) && \text{(factor out } n)
\end{aligned}
$$

Theorem 0.6 (Score function of a binary sample)**.** *If* $Y_i \overset{\perp\!\!\!\perp}{\sim} \text{Ber}(\pi)$ *for* $i = 1, \ldots, n$*, with observed sample mean* $\bar{y}$ *and precision* $\tau = 1/(\pi(1-\pi))$*, the score function is:*

$$\ell'(\pi) = n\tau(\bar{y} - \pi)$$

Proof

Proof. This is the solution to Exercise 0.6, worked in Solution 0.6. □

Solving the score equation

Exercise 0.7 (Solving the score equation of a binary sample)**.** In the setting of Exercise 0.6, assume $0 < \bar{y} < 1$: the sample has at least one event and at least one non-event. Solve the score equation $\ell'(\pi) = 0$ for $\pi \in (0, 1)$.

Solution

Solution 0.7. For $\pi \in (0, 1)$, $\tau = 1/(\pi(1-\pi))$ is finite and positive, and $n > 0$, so $n\tau \neq 0$.

$$\begin{aligned} 0 &= \ell'(\pi) && \text{(score equation)} \\ 0 &= n\tau(\bar{y} - \pi) && \text{(score function of a binary sample)} \\ 0 &= \bar{y} - \pi && \text{(divide both sides by } n\tau \neq 0) \\ \pi &= \bar{y} && \text{(add } \pi \text{ to both sides)} \end{aligned}$$

Since $0 < \bar{y} < 1$, this solution lies in $(0, 1)$.

Theorem 0.7 (Solution of the score equation of a binary sample)**.** *If* $Y_i \sim_{\perp\!\!\!\perp} \text{Ber}(\pi)$ *for* $i = 1, \ldots, n$ *and the observed sample mean satisfies* $0 < \bar{y} < 1$*, then the score equation* $\ell'(\pi) = 0$ *has exactly one solution in* $(0, 1)$*: the sample mean* $\pi = \bar{y}$*.*

Proof

Proof. This is the solution to Exercise 0.7, worked in Solution 0.7. □

Hessian contributions

Exercise 0.8 (Hessian contribution of one binary outcome)**.** In the setting of Exercise 0.5, derive the Hessian contribution $\ell_i''(\pi) \overset{\text{def}}{=} \frac{\partial}{\partial \pi}\ell_i'(\pi)$ of observation i. Start from the score contribution written as a difference of two fractions, $\ell_i'(\pi) = y_i/\pi - (1 - y_i)/(1 - \pi)$, the form in Solution 0.4 before its fractions are combined.

Solution

Solution 0.8.

$$\begin{aligned}
\ell_i''(\pi) &= \frac{\partial}{\partial \pi}\ell_i'(\pi) && \text{(definition of the Hessian contribution)}\\
&= \frac{\partial}{\partial \pi}\left(\frac{y_i}{\pi} - \frac{1-y_i}{1-\pi}\right) && \text{(score contribution as a difference of fractions)}\\
&= \frac{\partial}{\partial \pi}(y_i\pi^{-1} - (1-y_i)(1-\pi)^{-1}) && \text{(write each fraction as a negative power)}\\
&= y_i\frac{\partial}{\partial \pi}\pi^{-1} - (1-y_i)\frac{\partial}{\partial \pi}(1-\pi)^{-1} && \text{(linearity and constant multiple rule)}
\end{aligned}$$

The first derivative is:

$$\frac{\partial}{\partial \pi}\pi^{-1} = -\pi^{-2} \qquad \text{(power rule)}$$

The second derivative needs the chain rule:

$$\begin{aligned}
\frac{\partial}{\partial \pi}(1-\pi)^{-1} &= -(1-\pi)^{-2}\frac{\partial}{\partial \pi}(1-\pi) && \text{(chain rule and power rule)}\\
&= -(1-\pi)^{-2}(-1) && (\frac{\partial}{\partial \pi}(1-\pi) = -1, \text{ as in the score exercise})\\
&= (1-\pi)^{-2} && \text{(multiply)}
\end{aligned}$$

Substituting both derivatives back in:

$$\begin{aligned}
\ell_i''(\pi) &= y_i(-\pi^{-2}) - (1-y_i)(1-\pi)^{-2} && \text{(substitute the two derivatives)}\\
&= -\frac{y_i}{(\pi)^2} - \frac{1-y_i}{(1-\pi)^2} && \text{(write the negative powers as fractions)}
\end{aligned}$$

Theorem 0.8 (Hessian contribution of one binary outcome)**.** *If* $Y_i \sim \text{Ber}(\pi)$ *has observed value* $y_i \in \{0, 1\}$*, its Hessian contribution is:*

$$\ell_i''(\pi) = -\frac{y_i}{(\pi)^2} - \frac{1-y_i}{(1-\pi)^2}$$

Proof

Proof. This is the solution to Exercise 0.8, worked in Solution 0.8. □

The Hessian of the sample

Exercise 0.9 (Hessian of a binary sample)**.** In the setting of Exercise 0.6, assemble the Hessian $\ell''(\pi)$ of the whole sample from the Hessian contributions.

Solution

Solution 0.9.

$$
\begin{aligned}
\ell''(\pi) &= \sum_{i=1}^{n} \ell_i''(\pi) && \text{(the derivative of a sum is the sum of the derivatives)} \\
&= \sum_{i=1}^{n} \left(-\frac{y_i}{(\pi)^2} - \frac{1-y_i}{(1-\pi)^2} \right) && \text{(Hessian contribution of a binary outcome)} \\
&= -\frac{1}{(\pi)^2} \sum_{i=1}^{n} y_i - \frac{1}{(1-\pi)^2} \sum_{i=1}^{n} (1-y_i) && \text{(split the sum; the denominators do not depend on } i\text{)} \\
&= -\frac{n\bar{y}}{(\pi)^2} - \frac{1}{(1-\pi)^2}(n - n\bar{y}) && \left(\textstyle\sum_{i=1}^{n} y_i = n\bar{y} \text{ and } \sum_{i=1}^{n} 1 = n\right) \\
&= -\frac{n\bar{y}}{(\pi)^2} - \frac{n(1-\bar{y})}{(1-\pi)^2} && \text{(factor out } n\text{)} \\
&= -n \left(\frac{\bar{y}}{(\pi)^2} + \frac{1-\bar{y}}{(1-\pi)^2} \right) && \text{(factor out } -n\text{)}
\end{aligned}
$$

Theorem 0.9 (Hessian of a binary sample)**.** *If* $Y_i \overset{\perp\!\!\!\perp}{\sim} \text{Ber}(\pi)$ *for* $i = 1, \ldots, n$*, with observed sample mean* $\bar{y}$*, the Hessian of the log-likelihood is the* 1×1 *matrix whose only entry is:*

$$\ell''(\pi) = -n \left(\frac{\bar{y}}{(\pi)^2} + \frac{1-\bar{y}}{(1-\pi)^2} \right)$$

Proof

Proof. This is the solution to Exercise 0.9, worked in Solution 0.9. □

Checking that the Hessian is negative definite

Exercise 0.10 (Negative definiteness of the Hessian of a binary sample)**.** In the setting of Exercise 0.9, show that the Hessian $\ell''(\pi)$ is negative definite for every $\pi \in (0, 1)$.

Solution

Solution 0.10. A 1×1 matrix is negative definite exactly when its only entry is negative, so it is enough to show $\ell''(\pi) < 0$.
Fix $\pi \in (0, 1)$. Then:

- $(\pi)^2 > 0$ and $(1 - \pi)^2 > 0$;
- $\bar{y} \geq 0$ and $1 - \bar{y} \geq 0$, because $y_i \in \{0, 1\}$;
- $\bar{y}$ and $1 - \bar{y}$ are not both zero, because they sum to 1.

So $\bar{y}/(\pi)^2 + (1 - \bar{y})/(1 - \pi)^2$ is a sum of two nonnegative terms, at least one of them positive. It is therefore positive, and with $n > 0$:

$$\ell''(\pi) = -n\left(\frac{\bar{y}}{(\pi)^2} + \frac{1 - \bar{y}}{(1 - \pi)^2}\right) < 0$$

This holds for every $\pi \in (0, 1)$.

Theorem 0.10 (Negative definiteness of the Hessian of a binary sample)**.** *If* $Y_i \overset{\perp\!\!\!\perp}{\sim} \text{Ber}(\pi)$ *for* $i = 1, \ldots, n$*, the Hessian* $\ell''(\pi)$ *of the log-likelihood is negative definite for every* $\pi \in (0, 1)$*, whatever the observed outcomes. Hence the log-likelihood is strictly concave on* $(0, 1)$*.*

Proof

Proof. This is the solution to Exercise 0.10, worked in Solution 0.10. It builds on Theorem 0.9. □

The maximum likelihood estimate

Exercise 0.11 (Maximum likelihood estimate of a binary sample)**.** In the setting of Exercise 0.7, use Theorem 0.10 to find the maximum likelihood estimate $\hat{\pi}_{\text{ML}}$ of π.

Solution

Solution 0.11. By Theorem 0.10, ℓ is strictly concave on $(0,1)$. A strictly concave function that has a critical point is maximized there, and nowhere else. With $0 < \bar{y} < 1$, Theorem 0.7 says the only critical point of ℓ in $(0,1)$ is $\bar{y}$. So ℓ is maximized at $\pi = \bar{y}$.

Theorem 0.11 (Maximum likelihood estimate of a binary sample)**.** *If* $Y_i \overset{\perp\!\!\!\perp}{\sim} \text{Ber}(\pi)$ *for* $i = 1, \ldots, n$ *and the observed sample mean satisfies* $0 < \bar{y} < 1$*, the maximum likelihood estimate of* π *is the sample mean:*

$$\hat{\pi}_{ML} = \bar{y}$$

Proof

Proof. This is the solution to Exercise 0.11, worked in Solution 0.11. It builds on Theorem 0.7 and Theorem 0.10. □